

Project Team Organization – HR

Topic # 7

People (Human resources)

- The most important contribution in a software project is made not by the system or the tools but by the people.
- The human element is very vital in software project management.
- The success of a project depends on selecting the right kind of people with the right kind of talent.
- Depending on their roles and responsibilities, people involved in a software project can be categorized into the following main categories:
  - Senior manager
  - Project manager
  - Software Engineer
  - Customer
  - End user



Senior Managers



- The senior managers lay out the business policies and rules that affect the project or influence the project.
- They may or may not have the technical expertise required for the project.

Project Managers



- Project managers schedule, monitor and motivate the activities of the developers/programmers.
- They are required to have sound technical knowledge as well as managerial skills.
- They must have good leadership skills as they are expected to
  - motivate and organize the team that will eventually produce the software.

Software Engineers



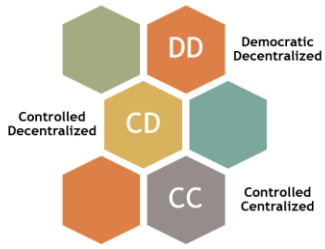
- These are the people who work “behind-the scenes”.
- Their roles can stretch from simple designing of the project to complex coding responsibilities.
- They need to have intense technical expertise.



Team Organization

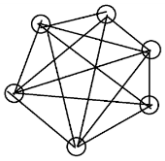
- If there are a number of people involved in a project, it is expected that they will be organized into one or more teams, each of which will be assigned certain tasks.
- These teams will then be coordinated by the teams themselves as well as the project manager.
- The best kind of team structure depends upon
  - the management style of an organization,
  - the number of people who will constitute the team,
  - their skill levels and
  - the overall complexity of the problem to be solved.

Broad Categories of Team Organizations



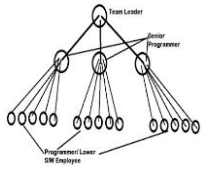
Democratic Decentralized (DD)

- If there are a number of people involved in a project, it is expected that they will be organized into one or more teams, each of which will be assigned certain tasks.
- This kind of team has a horizontal level of communication where there will be no permanent leader as such.
- A person to coordinate and monitor tasks will be appointed for short durations and then replaced by others who may coordinate different tasks.
- Decisions are generally made by group consent.
- This kind of team structure is best for difficult problems.
- Job satisfaction and high morale can be achieved by implementing this kind of team structure.



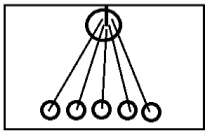
Controlled Decentralized (CD)

- This kind of team has a proper leader who coordinates specific tasks and secondary leaders who are responsible for subtasks.
- Problem solving is still a group activity but the team leader divides implementation of solutions among subgroups.
- Communication within subgroups and individuals is horizontal. However, along the control hierarchy, vertical communication also exists.
- This kind of team structure is suitable for simple problems and is found to produce fewer defects.



Controlled Centralized (CC)

- A team leader manages top-level problem solving and internal team coordination.
- Communication between the leader and team members is vertical.
- This kind of team structure is also suitable for simple problems and is found to produce fewer defects.



Factors considered while planning the structure of Software Engineering Teams

- The difficulty of the problem to be solved
- The time that the team will be needed to stay together
- The size of the resultant program(s) in lines of code or function points (*covered in detail in the later session*)
- The degree to which the problem can be broken up into modules
- The expected reliability and quality of the system to be built
- The expected amount of communication required for the project
- The rigidity of the delivery date

What structure can be used and when?

For solving simple problems	For solving difficult problems	For large projects	For high morale and job satisfaction	For solving problems with low modularity
• A centralized structure is more suitable since it completes faster than a decentralized one	• A decentralized structure is suitable since decentralized teams generate more and better solutions than individuals.	• Since the performance of a team is inversely proportional to the amount of communication that must be conducted, very large projects can be addressed by teams with a CC or CD structure where sub-grouping can be easily accommodated.	• The DD team structure results in high morale and job satisfaction and are therefore good for teams that will be together for a long time.	• The DD team structure is best applied to problems with relatively low modularity because of the higher volume of communication needed. • When high modularity is possible (and people can do their own thing), the CC or CD structure will work well.

- *CC and CD teams have been found to produce fewer defects than DD teams.*
- *Decentralized teams generally require more time to complete a project than a centralized structure and at the same time are best when high sociability is required.*

Project Roles

- **Programmers**
  - Technical Lead
  - Architect
  - Programmer
- **Quality Assurance Engineers**
  - QA Manager
  - QA Lead
  - QA Staff
- **DBAs**
  - DB Administrator
  - DB Programmers
  - DB Modeler
- **Network Engineers**
- **System Administrators**
- **Analyst & Business Analyst**
- **UI Designers**
- **Information Architects**
- **Documentation Writers**
  - Editors
  - Documentation Specialist
- **Project Manager**
- **Other**
  - Security Specialist
  - Consultants
  - Trainers

END OF TOPIC 7

- **COMING UP!!!!!!**
- **Resource Allocation**
- **Project Control**
- **Midterm Exam II**
- **Agile & User Stories**